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# E-Commerce Market Portal

*A Full-Stack Web Application for Online Product Sales and Order Management*

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|-------------------------|----------------------------------------|
| <b>Project Title</b>    | E-Commerce Market Portal               |
| <b>Subject</b>          | Web Technologies                       |
| <b>Project Type</b>     | Full-Stack Web Application             |
| <b>Difficulty Level</b> | Intermediate                           |
| <b>Semester</b>         | Semester 6, B.Tech / GSSIPU Curriculum |
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## Abstract

This paper presents the design and implementation of an E-Commerce Market Portal, a full-stack web application simulating the core functionality of online marketplaces such as Amazon, Flipkart, and Myntra. The system allows customers to register, browse a product catalog, manage a shopping cart, place orders, and track purchases, while administrators can add, update, and remove inventory. The application follows a standard three-tier web architecture: an HTML/CSS/JavaScript frontend, a PHP backend handling business logic and authentication, and a MySQL relational database persisting users, products, and orders. This paper documents the system architecture, database schema, representative frontend and backend code, the end-to-end purchase workflow, and the security considerations relevant to a login-based e-commerce system.

*Keywords: e-commerce, full-stack web development, PHP, MySQL, HTML, CSS, JavaScript, authentication, shopping cart, three-tier architecture*

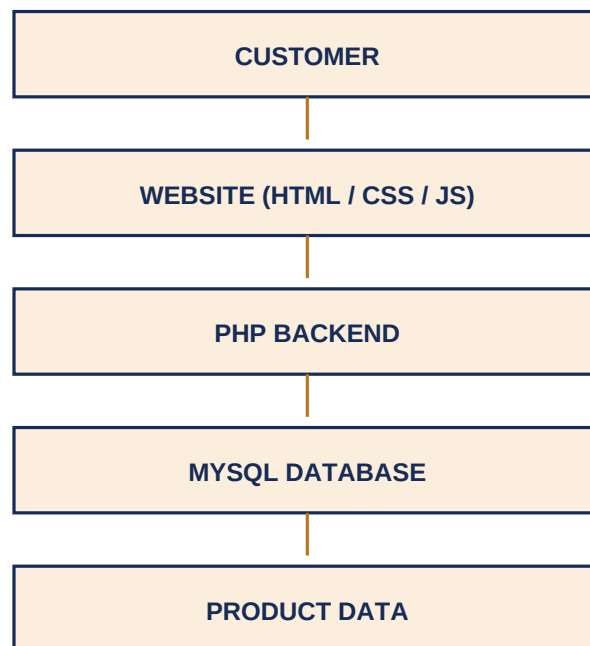
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## 1. Introduction

Online marketplaces have become the dominant channel for retail commerce, with platforms such as Amazon, Flipkart, and Myntra allowing customers to browse, purchase, and track products entirely through a web interface. This project, undertaken as a Semester 6 mini project under Web Technologies, builds a simplified but functionally complete E-Commerce Market Portal that reproduces this workflow: user registration and login, product browsing, cart management, order placement, and administrative inventory control. The project is intended to demonstrate practical, full-stack web development spanning client-side interactivity, server-side business logic, and relational data persistence.

## 2. System Architecture

The application follows a standard three-tier web architecture, illustrated in Figure 1. The customer interacts with the website's HTML, CSS, and JavaScript frontend, which renders product listings, the cart, and order status. User actions — login attempts, cart updates, checkout requests — are submitted to a PHP backend, which contains the application's business logic: validating credentials, computing order totals, and applying inventory changes. The PHP layer communicates with a MySQL database that persistently stores user accounts, the product catalog, and order records. This separation of concerns keeps presentation, business logic, and data storage independently maintainable.



*Figure 1. Three-tier architecture: customer, website frontend, PHP backend, and MySQL database.*

## 3. Main Modules

The system is divided into two primary modules. The User Module covers registration, login, product browsing, adding items to cart, placing orders, and tracking order status. The Admin Module covers adding new products,

updating existing product details, deleting discontinued products, and managing orders and inventory levels. Table 1 summarizes the database tables that back these modules.

| Table    | Fields                                              |
|----------|-----------------------------------------------------|
| Users    | UserID (PK), Name, Email, Password                  |
| Products | ProductID (PK), ProductName, Price, Quantity        |
| Orders   | OrderID (PK), UserID (FK), ProductID (FK), Quantity |

Table 1. Core database tables. PK = Primary Key, FK = Foreign Key.

## 4. Purchase Workflow

The end-to-end customer journey follows seven stages: registration, login, browsing the product catalog, adding selected products to the cart, proceeding to checkout, completing payment, and receiving order confirmation. Each stage corresponds to a distinct page in the frontend — home page, login page, product page, cart page, and checkout page — with the admin dashboard providing a separate interface for inventory management outside this customer-facing flow.

## 5. Frontend Implementation

The frontend uses standard HTML for page structure, CSS for visual styling, and JavaScript for client-side interactivity such as cart updates without a full page reload. Listing 1 shows a minimal HTML entry point linking to the product listing page.

```
<!DOCTYPE html>
<html>
<head>
<title>E-Commerce Portal</title>
</head>
<body>

<h1>Online Store</h1>
<a href="products.php">View Products</a>

</body>
</html>
```

Listing 1. Minimal HTML homepage linking to the PHP-rendered product listing.

## 6. Backend Authentication Logic

User login is handled server-side in PHP. The submitted email and password are used to query the Users table via a MySQLi connection; a non-empty result set indicates valid credentials. Listing 2 shows the core authentication query.

```
<?php
$conn = mysqli_connect("localhost", "root", "", "ecommerce");

$email = $_POST['email'];
```

```
$password = $_POST['password'];

$sql = "SELECT * FROM users
        WHERE email='$email' AND password='$password'";
$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {
    echo "Login Success";
} else {
    echo "Invalid Login";
}
?>
```

*Listing 2. PHP login handler querying the Users table via MySQLi.*

**Security note (evidence-supported):** concatenating raw POST input directly into a SQL string, as shown above, is vulnerable to SQL injection. A production system should use parameterized queries (prepared statements) and hash stored passwords (e.g. with *bcrypt*) rather than storing or comparing plaintext values. This listing reflects the introductory implementation used for this course project; Section 8 discusses the security hardening a deployed system would require.

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## 7. Database Schema in SQL

The Products table, central to the catalog and inventory features, is created with the following schema, using a decimal type for currency-accurate price storage.

```
CREATE TABLE Products (  
    ProductID INT PRIMARY KEY,  
    ProductName VARCHAR(100),  
    Price DECIMAL(10,2),  
    Quantity INT  
);
```

*Listing 3. SQL schema for the Products table.*

## 8. Security Features

Three security mechanisms are relevant to a login-based e-commerce system. Login authentication verifies a user's identity against stored credentials before granting access to account-specific features such as order history. Session management maintains a user's authenticated state across page requests using PHP sessions, avoiding the need to re-authenticate on every action. Database validation constrains inputs (e.g. via prepared statements and type checks) to prevent malformed or malicious records from being written to the database. A production deployment would additionally require HTTPS transport encryption, hashed password storage, and CSRF protection on state-changing form submissions.

## 9. Applications

The architecture and workflow demonstrated in this project generalize directly to online shopping platforms, fashion stores requiring inventory management, grocery platforms handling online orders, electronics stores maintaining product catalogs, and multi-vendor digital marketplaces.

## 10. Result

An E-Commerce Market Portal was successfully developed using HTML, CSS, JavaScript, PHP, and MySQL. The application supported product browsing, cart management, user authentication, and order processing, demonstrating full-stack web development across all three architectural tiers. The accompanying interactive HTML tool reproduces the complete user and admin workflow — login, browsing, cart, checkout, order tracking, and inventory management — against simulated in-memory data standing in for the PHP/MySQL backend described in this paper.

## References

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- GGSIPU B.Tech Computer Science Engineering Curriculum, Semester 6 — Web Technologies course structure.

— *Mayank Swaraj*